

Exercise : Caesar Cipher

1. Encrypt the following cipher text using caesar cipher with shift key of 3
plain text : CYBER SECURITY

Cipher letter	Shift (+3)
C	F
Y	B
B	E
E	H
H	K
K	N
N	Q
Q	T
T	W
W	Z

3. If two letters are in the same row, then replace with immediate right letter

Plaintext - dcd

P	L	A	Y	F	P	L	A	Y	F
I	J	K	L	M	I	J	K	L	M
N	O	P	Q	R	N	O	P	Q	R
S	T	U	V	W	S	T	U	V	W
X	Y	Z	A	B	X	Y	Z	A	B
C	D	E	F	G	C	D	E	F	G
H	I	J	K	L	H	I	J	K	L
M	N	O	P	Q	M	N	O	P	Q
R	S	T	U	V	R	S	T	U	V
W	X	Y	Z	A	W	X	Y	Z	A

Ex: plaintext - good morning

key - Playfair example

good morning - go/od /mo/or /in/gy

dq/voles/cu/rsc/gg

P	L	A	Y	F
I	J	K	L	M
N	O	P	Q	R
S	T	U	V	W
X	Y	Z	A	B
C	D	E	F	G
H	I	J	K	L
M	N	O	P	Q
R	S	T	U	V
W	X	Y	Z	A

ciphertext: dqvoescurk
dqvoescurk

Ex: 3

Prepare and encrypt - COMPUTER
Keyword - SECURITY

C O M P U
T E R A B
D F G H I
K L N O S
V W X Y Z

COMPUTER
OMPU CBRA
OMPUCBRA

SECURITY

SECUR
ITY A B
D F G H K
L M N O P
Q V W X Z

Plain text

COMPUTER
UN NL EA C
UNNLEAC

① Scenario: Event communication security.

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing a message 'MEET AT NOON' is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of encoding rule, would retrieve the original message.

5) Scenario: Legacy encryption in corporate communication.

A legacy system in a company still use a transposition-caused encryption method with three levels of arrangement. A received message appears as "WECRLTEERDS QEEFE AOC AIVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of message in this scenario.

(;) Vigenet cipher

Ex: Key: Raju

PT: welcome to cse department

$$(x+w) \bmod 26$$

$$(17+22) \bmod 26 = (P+K) \cdot 26$$

$$39 \cdot 26 - 13 = 11$$

PT: welcome to cse department.

CT: ne u w.

Railfence

* Used to transposition the character

* The PT is written as a sequence of diagonals of any c

* Then read them as a sequence of rows, that becomes a C

ex:

PT - meet me after the day

Row $\rightarrow$ diagonal depth is 2.

Example

- * PT-attack postponed until twam
- * Assume matrix column size is 7. That is optional

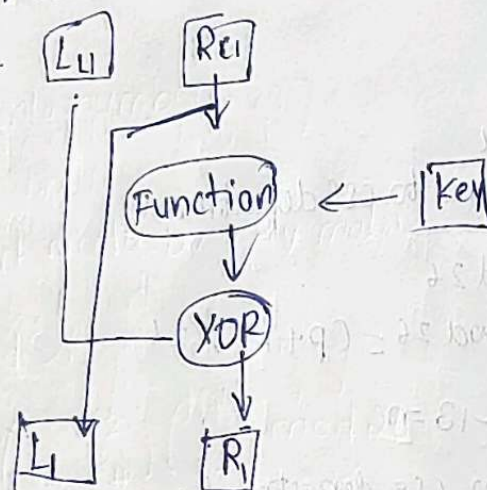
PT $\rightarrow$ $\begin{bmatrix} 4 & 3 & 1 & 2 & 5 & 6 & 7 \\ d & t & t & a & c & k & p \\ o & s & t & p & o & n & e \\ d & u & n & t & i & t & t \end{bmatrix}$

(ii) Fiestel cipher structure

- * Follows block cipher approach

Block cipher :

divide the text into different no. of blocks and perform operation on each block.



- ① A Fiestel cipher uses an 8-bit block and one round.

plain text = 10101101

Round key K_1 = 1100

Round function: $F(R, K) = R \oplus K$

Question

- * Split the plain text into L_0 & R_0
- * Compute

② Two round fistel cipher

* An 8-bit plaintext is encrypted using two fistel rounds

* plaintext = 11100010

* $K_1 = 0110$

* $K_2 = 1001$

③ Decryption Using fistel structure

* A cipher text 0011100 was produced by one round fistel cipher

* Round key $K_1 = 0101$

* $F(R, K) = R \oplus K$

Questions:

* Split the cipher text into L_1 & R_1

* Recover R_0 using the fistel property

* compute $F(R_0, K_1)$

* Recover the original L_0

* Write the recovered plaintext block

②

* Given,

plaintext = 11100010

$K_1 = 0110$

$K_2 = 1001$

* $L_0 = 1110$

$R_0 = 0010$

Round 1: $F(R_0, K_1) = 0010 \oplus 0110 = 0100$

$L_1 = R_0 = 0010$

$R_1 = L_0 \oplus F = 1110 \oplus 0100 = 1010$

Round 2: $F(R_1, K_2) = 1010 \oplus 1001 = 0011$

$L_2 = R_1 = 1010$

$R_2 = L_1 \oplus F = 0010 \oplus 0011 = 0001$

Final: 10100001

Round:	Left	Right
Initial	1110	0010
Round 1	0010	1010
Round 2	1010	0001

Answer: 10100001

(iv) Hill cipher.

- * 'Classical' polygraphic substitution cipher
- * It uses matrix multiplication to encrypt blocks of letters.

Steps:

- * Convert letters into numbers
- * Group letters into blocks
- * Multiply the block vector by a key matrix

$$C = KP \pmod{26}$$

$$(C_1 C_2) = \begin{pmatrix} K_1 & K_2 \\ K_3 & K_4 \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \pmod{26}$$

Step 1: $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Letter to number mapping

Letter step 1: $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

determinant check: $(3 \times 5) - (3 \times 2) = 15 - 6 = 9$

Since $\gcd(9, 26) = 1$, the matrix is valid.

Step 2: plain $K \times 4$

* HI

Convert to number

$$H=7$$

$$I=8$$

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Step 3: Multiply:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 49 \\ 54 \end{bmatrix}$$

Step 4: Take modulo 26:

$$\begin{bmatrix} 49 \\ 54 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 2 \end{bmatrix}$$

$$19 \rightarrow T$$

$$2 \rightarrow C$$

HELO

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

HE
LL
OX

$$P = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 33 \\ 34 \end{bmatrix}$$

(1) Take modulo 26: $33 \bmod 26 = 7$
 $34 \bmod 26 = 8$

$$7 \rightarrow H$$

$$8 \rightarrow I$$

(2) LL $P = \begin{bmatrix} 11 \\ 11 \end{bmatrix}$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 11 \\ 11 \end{bmatrix} = \begin{bmatrix} 66 \\ 77 \end{bmatrix}$$

modulo 26: $66 \bmod 26 = 14$
 $77 \bmod 26 = 25$

$$14 \rightarrow O$$

$$25 \rightarrow Z$$

(3) OX $P = \begin{bmatrix} 14 \\ 23 \end{bmatrix}$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 14 \\ 23 \end{bmatrix} = \begin{bmatrix} 111 \\ 143 \end{bmatrix}$$

$$111 \bmod 26 = 7$$

$$143 \bmod 26 = 17$$

Encrypted text: HIOZHN

Decryption for H10ZHN

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Step 1: $|K| = (3 \times 5) - (3 \times 2) = 15 - 6 = 9$

$$9 \times 3 = 27 \equiv 1 \pmod{26}$$

So, $q^{-1} = 3$

Matrix: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ Adjoint: $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$Adj(K) = \begin{bmatrix} 5 & -3 \\ 2 & 3 \end{bmatrix}$$

Convert negative no. to modulo 26

$$-3 \equiv 23$$

$$-2 \equiv 24$$

$$Adj(K) = \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix}$$

Inverse matrix

Multiply each element by 3

$$K^{-1} = 3 \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix} = \begin{bmatrix} 15 & 69 \\ 72 & 9 \end{bmatrix}$$

modulo 26:

$$69 \pmod{26} = 17$$

$$72 \pmod{26} = 20$$

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

Convert ciphertext to number

H1

OZ

HN

$$C = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 24 \\ 212 \end{bmatrix}$$

modulo 26: $24 \pmod{26} = 24$

$$212 \pmod{26} = 4$$

HE
E

② OZ $C = \begin{bmatrix} 14 \\ 28 \end{bmatrix}$

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 14 \\ 28 \end{bmatrix} = \begin{bmatrix} 635 \\ 505 \end{bmatrix}$$

$$635 \text{ modulo } 26 = 11$$

$$11 = L$$

$$505 \text{ mod } 26 = 11$$

$$11 = L$$

text = LL.

③ HN $C = \begin{bmatrix} 7 \\ 13 \end{bmatrix}$

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 13 \end{bmatrix} = \begin{bmatrix} 326 \\ 252 \end{bmatrix}$$

Modulo 26:

$$326 \text{ mod } 26 = 14$$

$$14 = O$$

$$252 \text{ mod } 26 = 23$$

$$23 = V$$

HELLO

~~text: OV~~

Decrypted text = HELLO

23/2